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Research and Applications

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Large language models are less effective at clinical prediction tasks than locally trained machine learning models

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Abstract

Objectives: To determine the extent to which current large language models (LLMs) can serve as substitutes for traditional machine learning (ML) as clinical predictors using data from electronic health records (EHRs), we investigated various factors that can impact their adoption, including overall performance, calibration, fairness, and resilience to privacy protections that reduce data fidelity.

Materials and Methods: We evaluated GPT-3.5, GPT-4, and traditional ML (as gradient-boosting trees) on clinical prediction tasks in EHR data from Vanderbilt University Medical Center (VUMC) and MIMIC IV. We measured predictive performance with area under the receiver operating characteristic (AUROC) and model calibration using Brier Score. To evaluate the impact of data privacy protections, we assessed AUROC when demographic variables are generalized. We evaluated algorithmic fairness using equalized odds and statistical parity across race, sex, and age of patients. We also considered the impact of using in-context learning by incorporating labeled examples within the prompt.

Results: Traditional ML [AUROC: 0.847, 0.894 (VUMC, MIMIC)] substantially outperformed GPT-3.5 (AUROC: 0.537, 0.517) and GPT-4 (AUROC: 0.629, 0.602) (with and without in-context learning) in predictive performance and output probability calibration [Brier Score (ML vs GPT-3.5 vs GPT-4): 0.134 vs 0.384 vs 0.251, 0.042 vs 0.06 vs 0.219)].

Discussion: Traditional ML is more robust than GPT-3.5 and GPT-4 in generalizing demographic information to protect privacy. GPT-4 is the fairest model according to our selected metrics but at the cost of poor model performance.

Conclusion: These findings suggest that non-fine-tuned LLMs are less effective and robust than locally trained ML for clinical prediction tasks, but they are improving across releases.

Key words: large language models; clinical prediction models; fairness; privacy.

Introduction

Making predictions is a vital part of healthcare, with common applications, including estimating the likelihood of a specific diagnosis,[1–3] assessing the suitability of a medication,[4,5] and determining a patient's readiness for discharge.[6] Over the past several decades, the practice of medicine has evolved from clinical intuition to incorporate data-driven guidance, which is increasingly aided by artificial intelligence (AI) and particularly machine learning (ML). While model development has advanced, it is widely recognized that implementing ML models successfully often requires training on or fine-tuning (eg, via transfer learning)[7] with local, representative data.[1,2]

However, not all institutions have sufficient resources to implement ML effectively. For instance, community hospitals may lack the volume of ML-ready data, computational power, and personnel that academic medical centers possess.[8] More recently developed ML technologies, such as large language models (LLMs), have hinted at a potential to mitigate these challenges and fundamentally change the integration of ML in medicine. Closed-source LLMs, such as GPT-3.5,[9] GPT-4,[10] and Claude,[11] are pre-trained and can be interacted with conversationally, characteristics that reduce technical friction required to create or use ML (or more broadly, AI) for healthcare settings. LLMs have been used to aid in various tasks, including drafting patient messages,[12] optimizing clinical decision support,[13] developing phenotyping algorithms,[14] and summarizing patient encounters.[15] There have been recent investigations into the performance[16–19] and fairness[20] of LLMs versus traditional ML and other diagnostic aids.[17,21] More recently, Chen et al[22] analyzed a wide range

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of LLMs for clinical prediction tasks; however, a number of critical aspects, including reliability and robustness (eg, model calibration) of these tools, remain unknown. Given the rapid pace at which LLMs are growing in their production,[23] it is critical to quantify their strengths and weaknesses to ensure responsible use.

To learn more about the efficacy of LLMs for clinical prediction, several aspects of real-world implementation need to be considered. First, we need to represent the type of users and constraints that may limit how LLMs are applied in a healthcare setting. In this study, we consider 2 distinct types of clinical users: (1) the everyday user, whose goal is to retrieve an answer using a single LLM query (ie, zero-shot prompting) and (2) the advanced user who utilizes retrieval augmented generation (RAG) to contextually present the LLM with similar cases or additional relevant examples in the prompt to augment its decision-making capabilities for better results. In-context learning is such a strategy that helps guide LLM inference.[16] We assume that the end-user is a clinical professional who will use the output of the LLM in support of their activities.[24] Clinicians are not expected to blindly accept the output of the LLM as their sole source of information in decision-making; however, analyzing the downstream effects of the LLM's prediction as a clinical decision support tool is outside the scope of this work.

Second, privacy concerns can limit the amount of patient information that can be supplied to an LLM under the control of a third party, such as OpenAI, as these technologies risk exposing protected health information (PHI) to organizations that are neither covered entities nor their business associates as required by the Health Insurance Portability and Accountability Act (HIPAA). Thus, it is likely that patient data would need to be de-identified before submission to the LLM. Moreover, while this may address the privacy requirements of federal regulations, such as the HIPAA Privacy Rule, evidence suggests that as data detail is reduced, the predictive performance of ML is weakened.[25] The impact on LLMs, however, remains unclear. Third, many commercial LLMs do not provide detailed documentation of their architecture or training data. As a result, reproducibility and the design of formal evaluation frameworks are limited for prediction tasks in healthcare.

In this study, we evaluate the utility, privacy, and fairness of LLMs compared to traditional ML, using electronic health record (EHR) data from Vanderbilt University Medical Center (VUMC)[6] to predict the likelihood of patient discharge from the hospital within 24 hours and the public-use MIMIC-IV and MIMIC-IV ED datasets from Beth Israel Deaconess Medical Center (BIDMC)[26,27] to predict the likelihood of transfer to the intensive care unit (ICU) within 24 hours after triage in the emergency department (ED). This work is structured as follows: "Materials and Methods" describes the datasets, LLMs, traditional ML implementation, and experimental methodologies for evaluating the performance and calibration, robustness to reduction in detail of a patient's demographic information, and algorithmic fairness. Then, the "Results" and "Discussion" sections describe the results and our discussion and their broader implications, respectively.

Materials and methods

This study was approved by VUMC's Institutional Review Board (IRB# 191892).

Datasets

This study relied on several EHR-derived datasets from VUMC and BIDMC. Table S2 provides distributions for the entire datasets and the LLM evaluation set. Missing categorical features are replaced with a "?". Missing numeric features were replaced with −1. We opted for this strategy because we assume that a clinician interfacing with an LLM may not use statistical imputation procedures. The MIMIC-IV and MIMIC-IV ED datasets ($n = 393\ 576$) contain hospital admission records and ED records, respectively, for BIDMC.[26–28] From these, we derived a dataset to predict if a patient will be transferred to an ICU within 24 hours of hospital admission based on their presentation during ED triage. In this dataset, the predictors include race, age, arrival transport type, triage vital signs, and the number of ED, ICU, and inpatient hospital stays for the past 30, 90, and 365 days.

The VUMC dataset ($n = 28\ 880$) was developed to predict the likelihood of patient discharge from the hospital within 24 hours.[6] We ran preliminary experiments using gradient-boosted trees implemented with CatBoost.[29] We predicted the likelihood of discharge using the demographic and audit log data features. We then calculated feature importance based on changes in the loss function (implemented within the CatBoost library) and selected the 15 most important audit log features. Those features and the demographic features—race, age, current day of week, insurance type, and Area Deprivation Index[30] of residence—were relied upon in this portion of our study.

GPT

Our experiments were run on a HIPAA-compliant Azure instantiation of OpenAI's Generative Pre-trained Transformer[10] (GPT) to guard PHI. We utilized GPT-3.5 and GPT-4, with temperature set to 0 (to reduce the likelihood of extraneous text being generated) and the maximum number of tokens set to 1024. We randomly selected 1500 data points from each dataset to query LLMs under a random sampling assumption to ensure results and claims are unbiased. The features for each dataset were represented in JSON. Table S2 specifies specific prompt structures for each dataset. We considered both zero-shot prompting (not providing additional examples) and few-shot examples (see below). We prompted the LLM to provide a prediction and a likelihood of correctness (Table S2) and used this output in our analysis.

Retrieval augmented generation

To implement RAG, we selected likely relevant examples that would inform LLM inference on the datapoint under consideration. To select instances that are likely to serve as useful in-context examples, we partitioned the dataset D into disjoint subsets D_1 and D_2. We opted for this partitioning scheme to create training and test sets from the entire 1500 LLM evaluation set and to minimize the possibility of data leakage in our RAG implementation. To determine the number of clusters, we measured the inertia (ie, the sum of squared distances of samples to their closest cluster center)[31] for each value of k, starting at 2 and continuing until the change in inertia was less than 0.001. We selected the value of k based on the Kneedle algorithm, which identified the point where inertia shows diminishing returns.[32] This defined clusters $C_1, C_2, \ldots, C_k$ for D_2. We performed k-means clustering to help ensure similar patients are selected as RAG

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examples for both D_1 and D_2. For each $x \in D_1$, we selected cluster C_i that is closest to x by Euclidean distance. We performed a sensitivity analysis (Figure S3) to determine the number of few-shot examples to use in our experiments. Our sensitivity analysis indicated that there was no noticeable advantage for selecting more than 5 in-context examples. Thus, we selected up to 5 data points in C_i and their corresponding labels as few-shot examples. If C_i had less than 5 examples, then we randomly selected additional data points in D_2 to ensure that there were 5 in-context examples. Otherwise, we selected the 5 points in C_i with the closest Euclidean distance to x (eg, 5-shot examples). We repeated this procedure to select few-shot examples for each $x \in D_2$ from D_1. We denote the use of in-context examples as GPT-X RAG Closest Within Cluster (or GPT-X RAG CWC).

Traditional ML

We used a traditional ML benchmark of a gradient-boosting tree (GBT) implemented with Catboost.[29] Categorical data were replaced with an integer corresponding to the index when sorting the possible values alphanumerically. We used default settings for the GBT and performed 10-fold cross-validation to collect the predictions for each data point per test set (ie, test-set predictions for each data point in dataset).

Statistical analysis

Unless otherwise noted, we produced a total of 30 bootstrapped samples of the predictions and report performance metrics as average and 95% confidence intervals. When applicable, we used Welch's t-test and 1-way analysis of variance (ANOVA) test and considered $P < .05$ as significant for all statistical tests. For receiver operating characteristic (ROC) curves and calibration curves, we performed vertical averaging to aggregate results from the bootstrap samples.[33] To perform vertical averaging of the graphs, we defined a comprehensive x-axis and averaged the corresponding y-values (interpolating y values if necessary). All analyses were performed using Python version 3.11.5, and the ANOVA test was calculated using SciPy library 1.11.4.

Design of performance and calibration experiments

To measure prediction performance, we used the area under the receiver operating characteristic (AUROC) curve. We performed vertical averaging[33] across the 30 bootstrap samples to construct the ROC curves. We compared the average AUROC using the bootstrapped samples (not the average AUROC from the vertically averaged curves since vertical averaging of curves interpolation of y values is required to maintain a consistent x axis).

Currently, accessing the LLM architecture or training data to assess model reliability or interpretability is not possible. Therefore, evaluating the reliability and applicability of predictions is crucial to ensuring interpretability and trustworthiness. As closed-source LLMs such as GPT-3.5 and GPT-4 are black box classifiers, their primary point of interaction is via prompt-based querying and the generated output. In our case, the generated output was the probability of a specific phenomenon (eg, likelihood of discharge, likelihood of ICU admittance).

In light of this, we used calibration curves[34] and Brier Score[35] to determine how well-calibrated output probabilities are from each model. Calibration curves plot the fraction of positive predictions given a predicted probability. A perfectly calibrated classifier should produce a calibration curve of $y = x$. This would indicate that, for all of the predictions assigned a predicted probability P, a fraction p of these predictions corresponds to the positive class. The Brier Score corresponds to the squared loss of predicted probabilities to output labels with a range of [0,1], where a lower value indicates a better-calibrated classifier—with 0 indicating perfect calibration and 1 indicating the worst possible calibration. The calibration curves were constructed by vertically averaging[33] calibration curves of the 30 bootstrap samples. The average Brier Score was calculated with the bootstrap samples, using the same technique employed with ROC curves.

We performed Welch's t-test to determine if there were statistically significant differences in both performance and calibration. Consider model_1 and model_2. The null hypothesis was $\mathcal{H}_0 : \mu_{\text{model}_1} = \mu_{\text{model}2}$. For AUROC, $\mathcal{H}_1 : \mu_{\text{model}_1} > \mu_{\text{model}_2}$, whereas for Brier Score, $\mathcal{H}_1 : \mu_{\text{model}_1} < \mu_{\text{model}_2}$. Thus, these hypothesis tests determined if model_1 performs better than model_2 for the appropriate metric.

Design of privacy-utility tradeoff experiments

To evaluate the privacy-utility tradeoff associated with LLMs, we considered the following patient data obfuscation procedure. Data were organized along 3 levels of detail as documented in Figure S3. Level 0 indicates no data obfuscation (ie, the original data); Level 1 indicates a mild amount of obfuscation, where each feature was divided into 4-8 categories; and Level 2 indicates moderate obfuscation where each feature was in a binary category (eg, for race, white/not white). Level 3 indicates all demographic data were suppressed (ie, removed entirely). Real-valued features were partitioned into ranges in Levels 1-3, while categorical features at Levels 1-3 were grouped into semantically similar categories. Sex was suppressed for Levels 1-3. To determine whether there was a statistically significant difference in LLM and ML performance between these levels, we performed a 1-way ANOVA test with a P-value threshold of .05.

Design of algorithmic fairness experiments

To assess the fairness of ML and LLMs, we considered subgroups (majority/minority) based on race (white/not white), age (18-49 years old/50 years or older), and sex (male/female). We measured algorithmic fairness in 2 ways: (1) average absolute odds difference (AAOD) and (2) statistical parity difference (SPD). AAOD is a measure based on the equalized odds (EO) measure of algorithmic fairness.[36] EO defines a fair classifier as one for which $\text{TPR}_{s_{\text{maj}}} = \text{TPR}_{s_{\text{min}}}$ and $\text{FPR}_{s_{\text{maj}}} = \text{FPR}_{s_{\text{min}}}$ for each possible pair of distinct demographic subgroups s_{maj} (majority) and s_{min} (minority). Then, we can calculate AAOD as follows: $\text{AAOD} = \frac{1}{2} |\text{FPR}_{s_{\text{maj}}} - \text{FPR } s_{\text{min}}| + \frac{1}{2} |\text{TPR}_{s_{\text{maj}}} - \text{TPR}_{s_{\text{min}}}|$. Thus, a perfectly fair classifier will have an AAOD of 0.

SPD measures the difference in the rate that each sensitive group receives a positive prediction, denoted as $\text{SPD} = P(\hat{Y} = 1 \mid A = s_{\text{maj}}) - P(\hat{Y} = 1 \mid A = s_{\text{min}})$.[37] An SPD of 0 corresponds to a perfectly fair classifier. A negative SPD indicates a preference for the minority class, whereas a positive SPD favors the majority class. Since SPD can range from −1 to 1, in addition to calculating the magnitude of statistical parity, we also calculated the absolute value of SPD (|SPD|). For Welch's t-test, we then have $\mathcal{H}_0 : \mu_{\text{model}_1} = \mu_{\text{model}2}$ and $\mathcal{H}_1 : \mu_{\text{model}_1} < \mu_{\text{model}_2}$. Since AAOD and |SPD| depict the

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magnitude of unfairness, smaller metric values indicate higher fairness. We report the sum of the fairness metrics per demographic subgroup.

Results

Performance and calibration

Table 1 summarizes the predictive performance of the LLMs and traditional ML model. As shown in Figure 1, the ROC curves for traditional ML were much larger than those for the LLMs, both with and without in-context learning. Moreover, the *P* values in Figure 2 indicated that this difference in performance between traditional ML and LLMs was statistically significant. Thus, in terms of measures of discrimination, traditional ML appeared to be more capable of identifying different classes of patients than LLMs at the given clinical tasks. This was not surprising because traditional ML learns a classification function based on a subset of representative data, and it has defined knowledge of expected feature distributions. In-context examples improved the ROC of GPT for all dataset-model combinations except for GPT-3.5 on VUMC data. Nonetheless, traditional ML still outperformed GPT with in-context examples.

Figure 3 depicts the calibration curves for traditional ML, GPT-3.5, and GPT-4. The classification curve for traditional ML nearly perfectly overlaid the $y = x$ curve. The calibration curves for GPT-3.5 and GPT-4, however, were heavily miscalibrated. These findings were corroborated by the Brier Score, which can be interpreted as a calibration loss, such that lower Brier scores imply better calibrated classifiers. Traditional ML was better calibrated than GPT-3.5 or GPT-4, a result that was statistically significant, as shown in Figure 2. In-context examples improved calibration for GPT-4 on both datasets, but GPT-3.5 exhibited a decrease in probability calibration for both datasets.

Privacy-utility tradeoff

Table 2 presents the mean performance (with 95% confidence intervals in parentheses) of each level of generalization per model. There was less than a 0.01 difference among the standard deviations of performance among different levels of generalization/suppression per model. The ANOVA analysis was statistically significant across all combinations of datasets and models (Table 3), indicating all models were affected, to some degree, by the generalization and suppression of demographic information.

Figure 4 displays the privacy-utility tradeoff across 4 levels of demographic generalization and suppression. We observed that traditional ML performance was generally consistent across the levels of demographic generalization and suppression, while GPT-3.5 and GPT-4 displayed varying degrees of performance but no overall directional trend. Next, we considered how LLMs compare to traditional ML as we reduced the granularity (generalize) and removed (suppress) demographic information. By visual inspection, traditional ML was relatively more robust to generalization and suppression than LLMs, and LLMs varied more in performance. There was no trend that was consistent across all evaluations. Generally, when we observed differences, less generalization appeared to lead to slightly better performance than full suppression. When demographic information was removed, LLMs (and particularly LLMs with in-context learning) dropped in performance. This raises the concern that LLMs may have some level of reliance on demographic information when performing zero-shot predictions.

Algorithmic fairness

Figures 5 and 6 plot model overall performance and fairness on an XY-plane. It was observed that traditional ML was not unambiguously the fairest model. For the VUMC data, GPT-4 with in-context examples was the fairest model across both fairness metrics, while in-context examples did not improve GPT-3.5 fairness. For the MIMIC dataset, GPT-4 was the fairest model, but in-context examples improved the fairness of GPT-3.5.

We further evaluated the statistical significance of differences in fairness metrics between different models by subgroup. Figure S4 provides heatmaps of *P* values from the corresponding Welch's *t*-tests determining the statistical significance of the results. There are 12 possible scenarios where one model could be statistically significantly fairer than another (based on demographic variable, fairness metric, and dataset). For all 12 cases, GPT-4 was statistically significantly fairer than traditional ML; however, incorporating in-context examples reduced this number to 10 of 12 cases. For GPT-3.5, zero-shot learning was statistically significantly fairer than traditional ML for only 6 of 12 cases. In the VUMC dataset, GPT-3.5 struggled with maintaining fairness for race, but in the MIMIC dataset, GPT-3.5 was only fairer for subgroups defined by sex. In-context learning did not help the fairness of the VUMC data; however, in-context examples improved GPT-3.5's fairness for the race and age variables in the mimic dataset. When comparing GPT-3.5 and GPT-4, we note that there is no unambiguously fairer model. Without in-context examples, GPT-4 surpassed GPT-3.5's fairness for all demographics in the MIMIC dataset but none in the VUMC dataset. With in-context examples, GPT-4 was fairer on the VUMC dataset, yet GPT-3.5 still surpassed GPT-4's fairness on the MIMIC dataset. We note that this fine-grained analysis may differ somewhat from the

Table 1. Average AUROC and Brier Scores and 95% confidence intervals.

	VUMC discharge		MIMIC-IV ED	
	AUROC ↑	Brier Score ↓	AUROC ↑	Brier Score ↓
GPT-3.5	0.537 (0.532, 0.543)	0.384 (0.38, 0.388)	0.517 (0.513, 0.521)	0.06 (0.057, 0.062)
GPT-4	0.629 (0.625, 0.634)	0.251 (0.249, 0.253)	0.602 (0.592, 0.612)	0.219 (0.216, 0.222)
GPT-3.5 RAG CWC	0.512 (0.509, 0.515)	0.477 (0.47, 0.484)	0.608 (0.596, 0.619)	0.078 (0.076, 0.081)
GPT-4 RAG CWC	0.638 (0.632, 0.644)	0.246 (0.244, 0.248)	0.61 (0.598, 0.621)	0.152 (0.15, 0.153)
Traditional ML	**0.847 (0.845, 0.848)**	**0.134 (0.133, 0.134)**	**0.894 (0.894, 0.895)**	**0.042 (0.042, 0.042)**

Arrow indicates a better direction. Best in bold (↑ = higher is better; ↓ = lower is better).
Abbreviations: AUROC: area under the receiver operating characteristic; ML: machine learning; VUMC: Vanderbilt University Medical Center.

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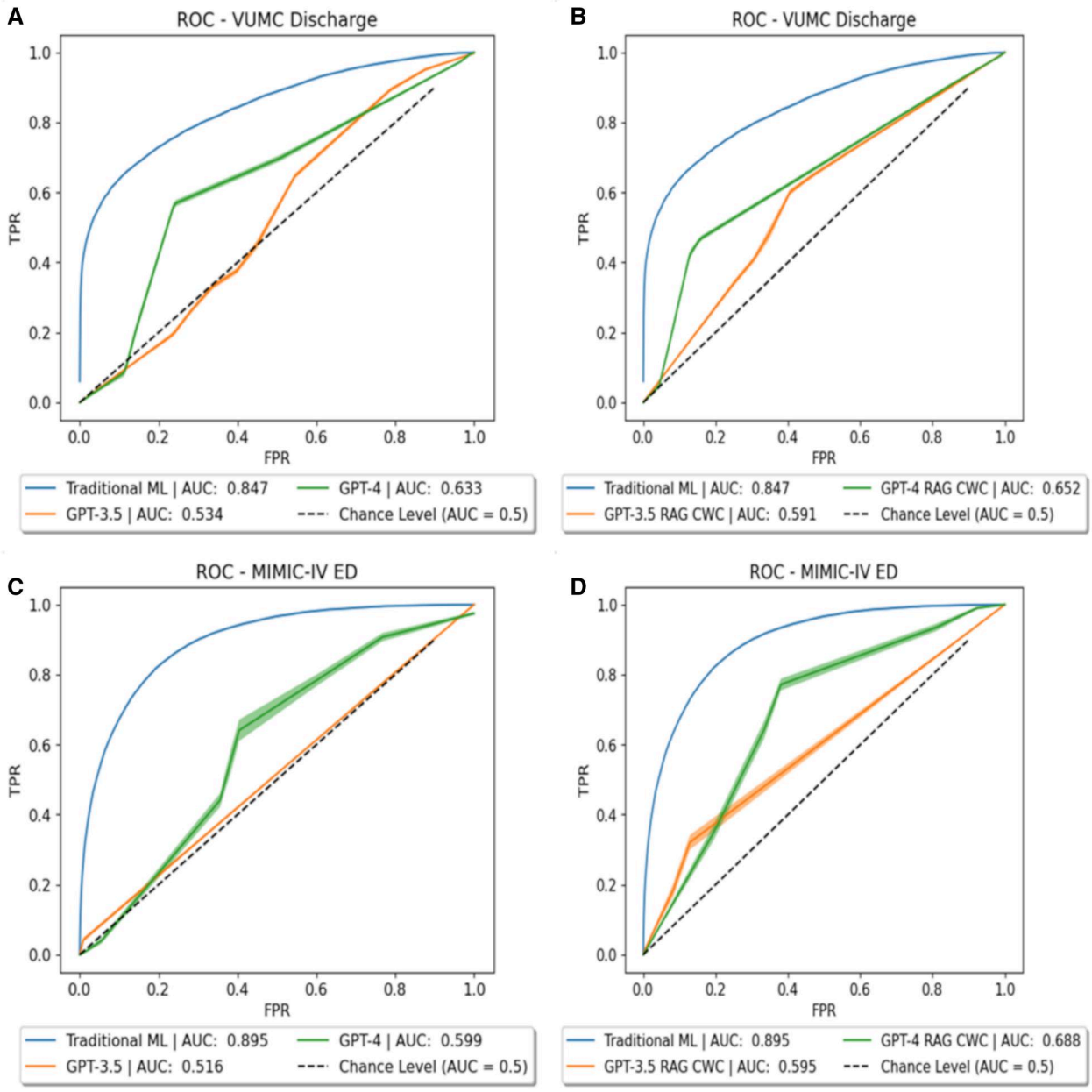


Figure 1. Receiver operating characteristic curves for (A) zero-shot prompting on the Vanderbilt University Medical Center (VUMC) dataset, (B) retrieval augmented generation (RAG)-based few-shot prompting on the VUMC dataset, (C) zero-shot prompting on the MIMIC-IV dataset, and (D) RAG-based few-show prompting on the MIMIC-IV dataset. Note that the AUC given in the legends may not match those given in Table 1 due to numerical interpolation of the curves required for vertical curve averaging (AUC is of the final averaged curve).

overall trends presented in Figures 5 and 6 since these figures consider aggregate fairness across all demographic subgroups considered. As a result, the small yet significant shifts may be canceled out in the aggregate.

While GPT-4 was the fairest model for both datasets, the fairer model between GPT-3.5 and traditional ML depended upon the dataset and fairness metric considered. Traditional ML was trained and evaluated on data from the same location. Thus, whatever biases are present in the data source will be present in the resulting model. We evaluated GPT-3.5 and GPT-4 using zero-shot prompting and without knowledge of the evaluation datasets. Thus, these models were not exposed to the local dataset biases that may influence decisions. This led us to hypothesize that traditional ML would be the most unfair model, which was validated by our findings. The use of in-context examples also impacted fairness; however, this impact was not unanimously positive or negative but depended upon dataset and metric. Moreover, despite being the fairer models, GPT-3.5 and GPT-4 were also the least capable predictors. Thus, it seems that overall improved fairness comes at the cost of reduced subgroup performance, but future work is necessary to confirm.

Discussion

This investigation suggests that LLMs are not yet ready to serve as predictive, analytic models, although GPT-4 does surpass GPT-3.5 in most of our experiments, indicating that

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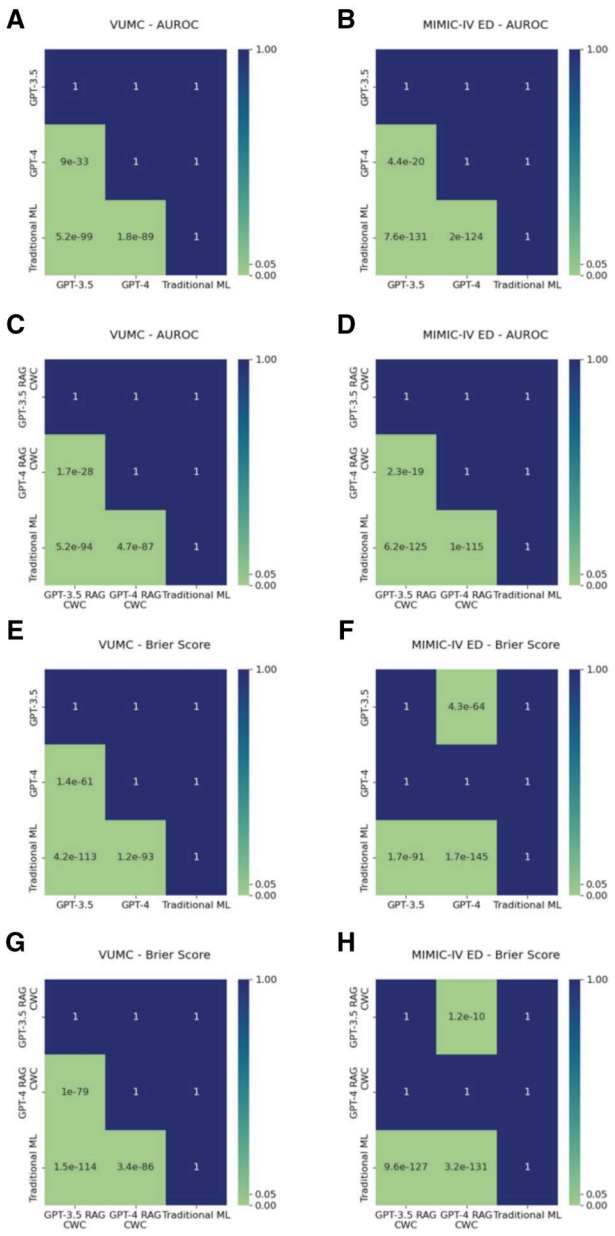


Figure 2. Let model_1 be the model given by the row and model_2 be the model given by the column. Then, $\mathcal{H}_0 : \mu_{\text{model}_1} = \mu_{\text{model}_2}$. For area under the receiver operating characteristic (AUROC) $\mathcal{H}_1 : \mu_{\text{model}_1} > \mu_{\text{model}_2}$. For Brier Score, $\mathcal{H}_1 : \mu_{\text{model}_1} < \mu_{\text{model}_2}$ Thus, these hypothesis tests determine if model_1 performs *better* than model_2 for the appropriate metric. (A) AUROC from zero-shot prompting on Vanderbilt University Medical Center (VUMC) dataset, (B) AUROC from zero-shot prompting on MIMIC-IV dataset, (C) AUROC from retrieval augmented generation (RAG)-based few-shot prompting on VUMC dataset, (D) AUROC from RAG-based few-shot prompting on MIMIC-IV dataset, (E) Brier Score from zero-shot prompting on VUMC dataset, (F) Brier Score from zero-shot prompting on MIMIC-IV dataset, (G) Brier Score from RAG-based few-shot prompting on VUMC dataset, (H) Brier Score from RAG-based few-shot prompting on MIMIC-IV dataset.

GPT is improving with new releases. In our experiments, LLMs have a significantly lower AUROC compared to traditional ML. This implies that LLMs simply do not match the discriminative capabilities of traditional ML for these tasks. These results are in line with a similar analysis recently performed by Chen et al.[22] Moreover, inspecting the calibration curves and Brier Scores reveals that LLMs are poorly calibrated when compared to the traditional ML comparison. The weak calibration of the LLM predictions in clinical scenarios calls into question their utility and reliability as a clinical classifier. This is not to say LLMs do not have advantages as a classifier, as there is little model tuning required out-of-the-box to be minimally functional (whereas traditional ML requires training data and computation time), and the conversational interaction may allow for chain-of-thought reasoning to support usability and interpretability.[38] Although further fine-tuning would likely be beneficial, our analysis assumes a resource-constrained environment where fine-tuning may not be feasible with respect to computational, data, and/or financial resources. These advantages, however, are difficult to realize when the output probabilities do not indicate reliability. Unlike several traditional ML algorithms,[39,40] LLMs have limited capability to quantify their uncertainty post hoc.[41] Thus, if LLMs are going to be useful as accurate clinical predictors, further research into improving the reliability and interpretability of LLMs is a must.

For LLMs to be useful as classifiers in lower resource settings, for example, where extensive data may not be available to train models, strategies such as in-context learning and RAG are necessary to produce good performance. We expected the incorporation of intelligently selected in-context examples[16,42] would improve LLM performance to levels near that of traditional ML. Our results suggest RAG does not help; however, we used what we consider a straightforward approach to RAG. There is emerging evidence that RAG-based examples are not always effective.[16,43] It is a reasonable conjecture that poor examples provided to an LLM are likely to result in a degradation in performance. There might be other approaches that would result in higher gains in performance; however, selection of an optimal RAG strategy is outside the scope of this work. Table 1 reveals that for the VUMC data, the usage of in-context examples resulted in either a slight increase in performance (GPT-4) or a decrease in performance (GPT-3.5). Further research is needed to improve the selection of RAG-based in-context examples for biomedical classification.

In the event that LLMs are the most accessible or applicable model for a situation, there are still concerns with respect to patient privacy when using these models. As we alluded to, organizations may apply some form of de-identification before sending data to the LLM. In this work, we evaluated the generalization of demographic PHI as well as suppression (denoted as Level 3 generalization/suppression in the results). We did not use a randomization strategy in order to maintain truthful privacy in line with the assumption that healthcare organizations do not want to falsify information about patients for predictive inference.[44–46] Our analysis indicates that LLMs are much more sensitive to the level of detail in demographic information compared to traditional ML. This suggests that a strategy that generalizes patient information may result in a reduction in predictive performance.

The apparent sensitivity of LLMs to the completeness of patient demographic information raises questions about the biases and unfairness present in these models based on their existing training. We find that GPT-4 is the fairest model evaluated; however, we note that LLMs are generally the lowest-performing models evaluated. By definition, an unfair model is one that results in a disparate outcome for at least one subgroup.[47] Typically, mitigating fairness concerns usually means improving the performance of the disadvantaged

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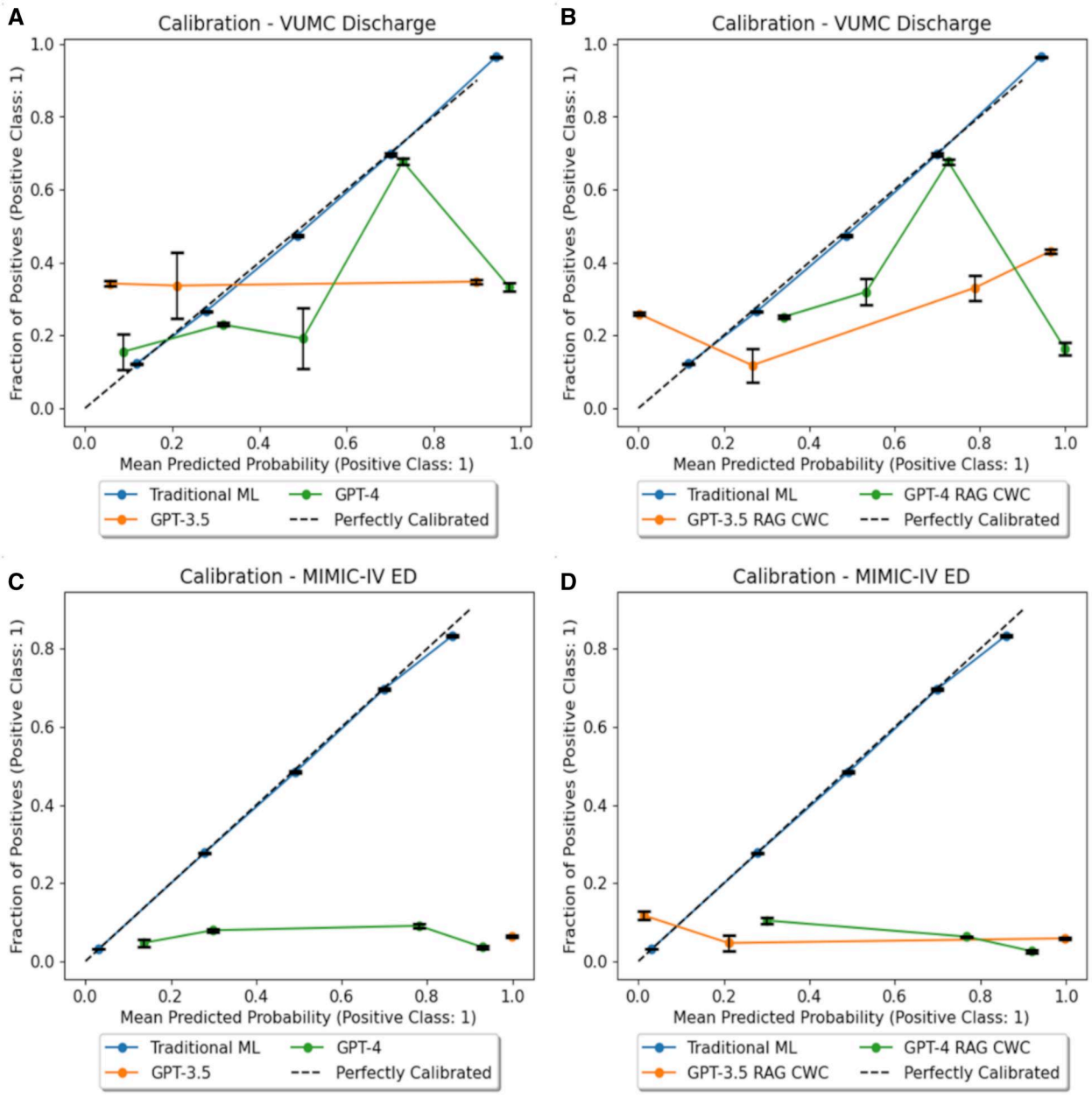


Figure 3. Calibration plots (A) zero-shot prompting on the Vanderbilt University Medical Center (VUMC) dataset, (B) retrieval augmented generation (RAG)-based few-shot prompting on the VUMC dataset, (C) zero-shot prompting on the MIMIC-IV dataset, and (D) RAG-based few-show prompting on the MIMIC-IV dataset.

subgroup(s). This route ensures equity of performance with reduced harm.[48] For the LLMs, increased fairness seems to come at reduced performance for all subgroups. This is evidenced by the overall degradation of performance of LLMs compared to traditional ML. Thus, it may be difficult to justify the use of LLMs on the basis of apparent fairness in the context of subpar performance.

Moreover, we find that there is not a consistently fair model. Fairness is conditioned on dataset and prediction task. For example, work from Liu et al[20] also compared the fairness of GPT-3.5 to traditional ML. In their analysis, traditional ML was fairer in their analysis when compared to GPT-3.5 in 2 of the 3 evaluated datasets. In our evaluation, traditional ML is fairer than GPT-3.5 for age and gender subgroups in the MIMIC dataset; however, this does not hold for the VUMC dataset or race in the MIMIC dataset. Liu et al also found that fine-tuning using few-shot examples improves the fairness gap for GPT-3.5, but there is little discussion on the performance impact. Our analysis indicates that few-shot examples can improve fairness of GPT-3.5 and GPT-4, but whether or not this occurs is dependent on model and dataset.

Though we conducted a large number of experiments, we acknowledge several limitations of this investigation. First, other facets of evaluation beyond the scope of this work, such as interpretability, were not evaluated and could also differ between model types. Second, we did not consider the costs of implementing and maintaining these models and the

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Table 2. Average AUROC and 95% confidence intervals for each model and level of data obfuscation.

		VUMC	MIMIC-IV ED
GPT-3.5	Level 0	0.537 (0.532, 0.543)	0.517 (0.513, 0.521)
	Level 1	0.573 (0.568, 0.578)	0.526 (0.522, 0.53)
	Level 2	0.575 (0.57, 0.579)	0.515 (0.511, 0.518)
	Level 3	0.557 (0.553, 0.561)	0.514 (0.514, 0.517)
GPT-4	Level 0	0.629 (0.625, 0.634)	0.602 (0.592, 0.612)
	Level 1	0.657 (0.652, 0.662)	0.633 (0.623, 0.644)
	Level 2	0.64 (0.636, 0.644)	0.609 (0.599, 0.619)
	Level 3	0.649 (0.644, 0.655)	0.601 (0.59, 0.611)
GPT-3.5 RAG CWC	Level 0	0.512 (0.509, 0.515)	0.608 (0.596, 0.619)
	Level 1	0.557 (0.551, 0.562)	0.631 (0.621, 0.642)
	Level 2	0.543 (0.54, 0.546)	0.605 (0.594, 0.615)
	Level 3	0.522 (0.516, 0.527)	0.564 (0.553, 0.575)
GPT-4 RAG CWC	Level 0	0.638 (0.632, 0.644)	0.61 (0.598, 0.621)
	Level 1	0.61 (0.605, 0.615)	0.617 (0.602, 0.631)
	Level 2	0.603 (0.598, 0.609)	0.62 (0.611, 0.629)
	Level 3	0.605 (0.599, 0.611)	0.53 (0.522, 0.538)
Traditional ML	Level 0	0.847 (0.845, 0.848)	0.894 (0.894, 0.895)
	Level 1	0.844 (0.843, 0.845)	0.894 (0.894, 0.894)
	Level 2	0.845 (0.844, 0.846)	0.892 (0.891, 0.892)
	Level 3	0.846 (0.844, 0.847)	0.894 (0.894, 0.895)

Abbreviations: AUROC: area under the receiver operating characteristic; ML: machine learning; VUMC: Vanderbilt University Medical Center.

Table 3. Results of 1-way ANOVA analysis to compare the effect of data generalization/suppression on LLM AUROC.

	GPT-3.5		GPT-3.5 RAG		GPT-4		GPT-4 RAG		Traditional ML	
Dataset	*F*	*P*	*F*	*P*	*F*	*P*	*F*	*P*	*F*	*P*
VUMC	48.341	2.42E−20	101.859	2.31E−32	45.743	1.39E−19	40.065	7.63E−18	17.441	2.92E−10
MIMIC	10.394	4.13E−06	50.665	5.29E−21	11.696	9.43E−07	96.963	1.80E−31	92.853	1.03E−39

Abbreviations: ANOVA: analysis of variance; AUROC: area under the receiver operating characteristic; LLM: large language model; ML: machine learning; VUMC: Vanderbilt University Medical Center.

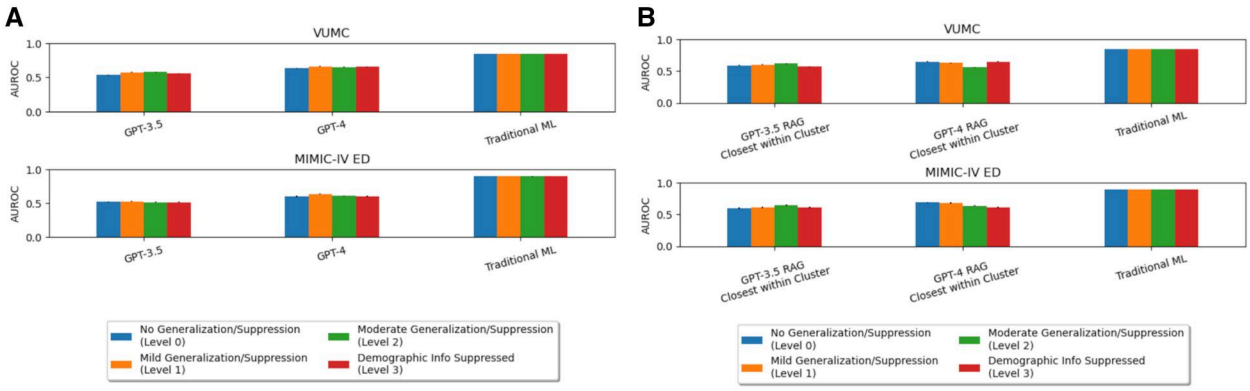


Figure 4. Bar charts depicting the privacy-utility tradeoff for (A) zero-shot prompting and (B) retrieval augmented generation (RAG)-based few-shot prompting.

associated environmental factors. Operational costs of AI may prohibit lower-resourced medical centers from utilizing these advances, and the energy required may have a downstream negative impact both economically and environmentally. Third, we only consider 2 clinical tasks from datasets derived from 2 medical centers. Additional studies that consider a wider variety of tasks across multiple medical centers are crucial for generalizability of these findings. Finally, we note that the privacy-utility tradeoff reported does not match the tradeoff expected. Ultimately, this calls into question the reliability of the datasets used for this privacy-utility tradeoff analysis. We cannot rule out the possibility that information captured in other predictors in these datasets are serving as appropriate proxies to demographic information.

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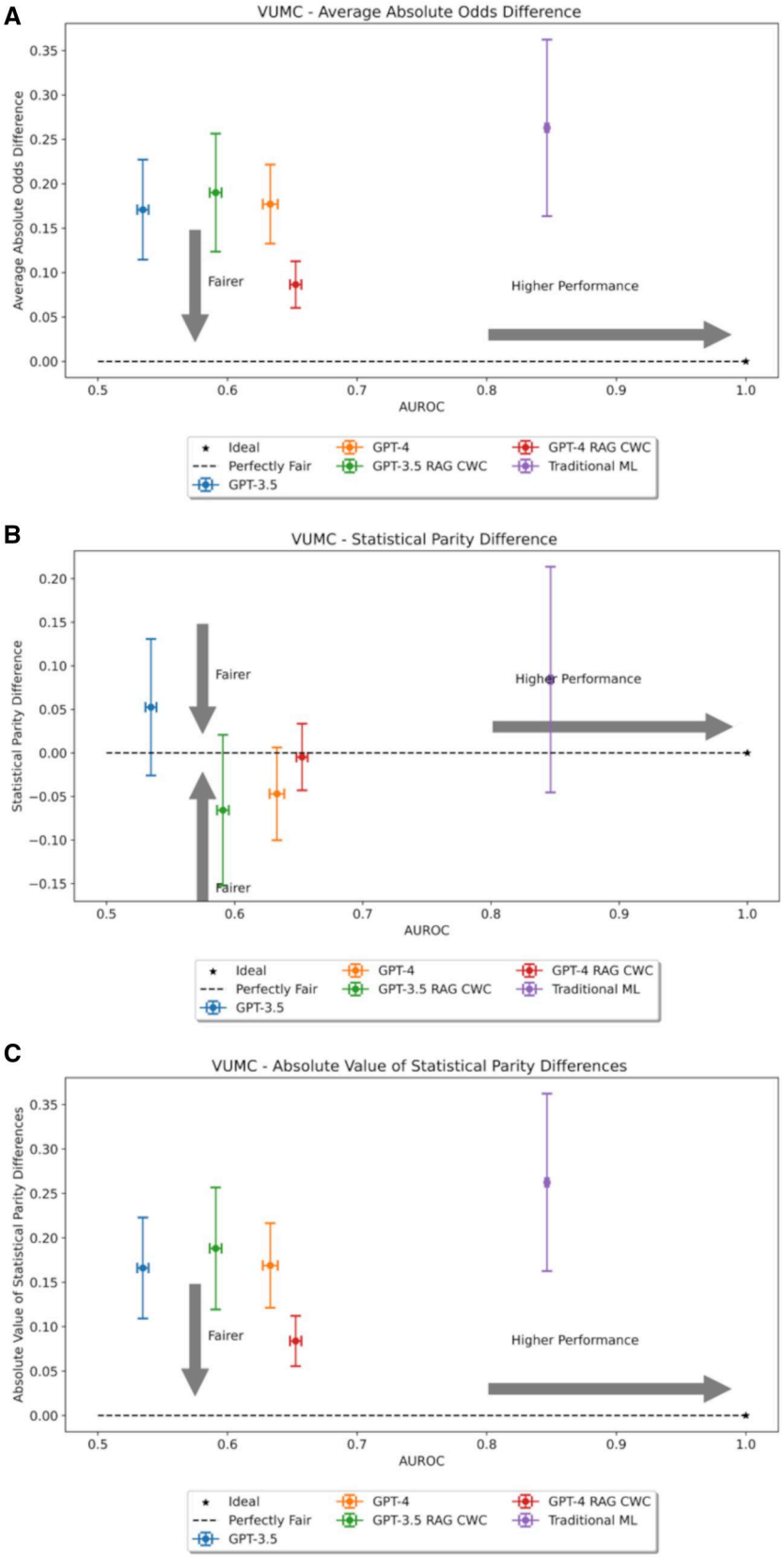


Figure 5. Two-dimensional fairness plots for the Vanderbilt University Medical Center (VUMC) dataset using (A) average absolute odds difference (AAOD), (B) statistical parity difference (SPD), or (C) absolute value of SPD as the fairness metric, respectively.

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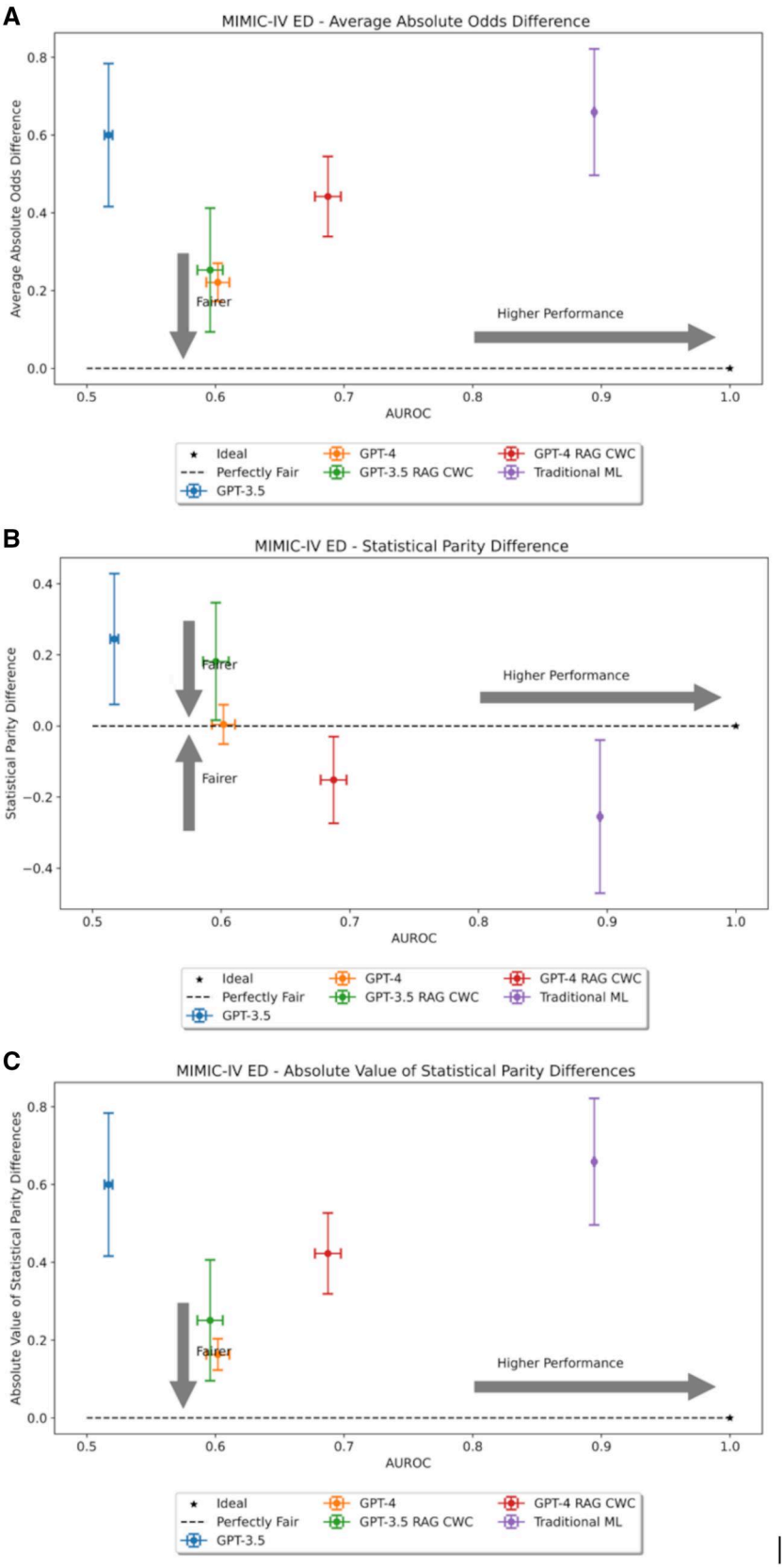


Figure 6. Two-dimensional fairness plots for the MIMIC-IV dataset using (A) average absolute odds difference (AAOD), (B) statistical parity difference (SPD), or (C) absolute value of SPD as the fairness metric, respectively.

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Conclusion

This study reported on a multifaceted comparison of LLMs to traditional ML in terms of model performance, calibration, privacy, and fairness as the points of evaluation. The evaluation of LLMs for clinical prediction tasks is a growing area of research[22] and as such, these benchmarks and analyses are still developing. We focused on 2 clinical prediction tasks: likelihood of ICU admission within 24 hours of ED arrival and likelihood of patient discharge from in-patient hospitalization within 24 hours. Even when supplemented with in-context learning, we find that LLMs are not as effective as traditional ML for clinical prediction tasks due to their poor discriminative performance and lack of reliability, particularly when demographic factors are variables in the prediction model. Our analysis was limited to 2 clinical prediction tasks, possibly limiting the generalizability of our claims. While LLMs are the fairest technologies, they are also those with poor model performance. This indicates that performance across subgroups degrades to ensure fairness, but this needs to be confirmed.

Still, we note that LLMs have potential. Since March 2023, LLMs have been used to aid with drafting patient messages,[12] optimizing clinical decision support,[13] developing phenotyping algorithms,[14] and summarizing patient encounters.[15] To reach this goal, research is needed to optimize in-context learning for LLM performance, understand the effect of randomization-based privacy preservation strategies, and understand the internal biases of LLMs to mitigate fairness concerns.

Author contributions

Katherine E. Brown (Conceptualization, Formal analysis, Investigation), Chao Yan (Conceptualization), Zhuohang Li (Writing), Xinmeng Zhang (Data curation), Benjamin X. Collins (Writing), You Chen (Data curation), Ellen Wright Clayton (Writing), Murat Kantarcioglu (Conceptualization), Yevgeniy Vorobeychik (Conceptualization), and Bradley A. Malin (Conceptualization, Resources, Supervision)

Supplementary material

Supplementary material is available at *Journal of the American Medical Informatics Association* online.

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Conflicts of interest

The authors have no conflicts of interest to disclose.

Data availability

The MIMIC-IV dataset underlying this article is available in Physionet at https://doi.org/10.13026/5ntk-km72. The VUMC data underlying this article cannot be shared publicly to preserve the privacy of individuals in the study. The data are available from the corresponding authors upon reasonable request.

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